ISLAMIC UNIVERSITY OF TECHNOLOGY

DHAKA, BANGLADESH

ORGANIZATION OF ISLAMIC COOPERATION

DATABASE MANAGEMENT SYSTEMS LAB

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Report preparing for Lab Task

Here is the problem statement -

Execute the banking.sql script using command. It creates a set of tables along with values that maintain the following schema:

ACTOR

ACT_ID NUMBER

ACT_FIRSTNAME
VARCHAR2

ACT_LASTNAME
VARCHAR2

ACT_GENDER

CASTS

ACT_ID NUMBER

MOV_ID NUMBER

ROLE VARCHAR2

REVIEWER

REV_ID NUMBER

REV_NAME VARCHAR2

DIRECTOR

DIR_ID NUMBER

DIR_FIRSTNAME
VARCHAR2

DIR_LASTNAME
VARCHAR2

MOVIE

MOV_ID NUMBER

MOV_TITLE VARCHAR2

MOV_YEAR NUMBER

MOV_LANGUAGE
VARCHAR2

MOV_RELEASEDATE
DATE

MOV_COUNTRY
VARCHAR2

RATING

MOV_ID NUMBER

REV_ID NUMBER

REV_STARS NUMBER

DIRECTION

DIR_ID NUMBER

MOV_ID NUMBER

Here, the boldfaces denote the primary keys and the arcs denote the foreign key relationships.

In this lab, you have to write all SQL statements in an editor first and save them with .sql extension.

Then execute the SQL script.

Write SQL statements for the following queries:

1.Find the name of the actors/actresses that are also directors (with and without set operator).

Solution Approach

```
SELECT DISTINCT ACT_FIRSTNAME, ACT_LASTNAME
FROM ACTOR
WHERE (ACT_FIRSTNAME || ' ' || ACT_LASTNAME) IN (
    SELECT DIR_FIRSTNAME || ' ' || DIR_LASTNAME
    FROM DIRECTOR
);

SELECT NAME
FROM (
    SELECT ACT_FIRSTNAME || ' ' || ACT_LASTNAME AS
NAME
    FROM ACTOR
    UNION ALL
    SELECT DIR_FIRSTNAME || ' ' || DIR_LASTNAME AS
NAME
    FROM DIRECTOR
)
GROUP BY NAME
HAVING COUNT(*) > 1;
```

SQL Command

```
ACT_FIRSTNAME        ACT_LASTNAME
-------------------- --------------------
Woody                Allen
Orson                Welles
Kevin                Spacey
```

2.Find the actresses with the same first name.

Solution Approach

```
SELECT ACT_FIRSTNAME, COUNT(*) AS NUM_ACTRESSES
FROM ACTOR
WHERE ACT_GENDER = 'F'
GROUP BY ACT_FIRSTNAME
HAVING COUNT(*) > 1;
```

SQL Command

```
ACT_FIRSTNAME        NUM_ACTRESSES
-------------------- -------------
Kate                             2
Jennifer                         2
```

3.Find the list of all the full names stored in the database.

Solution Approach

```
SELECT ACT_FIRSTNAME || ' ' || ACT_LASTNAME AS
FULL_NAME
FROM ACTOR
UNION
```

```sql
SELECT DIR_FIRSTNAME || ' ' || DIR_LASTNAME AS
FULL_NAME
FROM DIRECTOR;
```

SQL Command

```
FULL_NAME
----------------------------------------
Al Pacino
Alfred Hitchcock
Ali Astin
Andrei Tarkovsky
Bryan Singer
Christian Bale
Christopher Nolan
Claire Danes
Danny Boyle
David Aston
David Lean

FULL_NAME
----------------------------------------
Deborah Kerr
Dev Patel
Eddie Redmayne
Ewan McGregor
F. Murray Abraham
Felicity Jones
Frank Darabont
George Raft
Gus Van Sant
Harrison Ford
Hayao Miyazaki

FULL_NAME
----------------------------------------
Jack Clayton
Jack Nicholson
Jackie Chan
```

```
James Cameron
James Marsh
James Stewart
Jennifer Aniston
Jennifer Garner
John Boorman
Jon Voight
Kate Mara

FULL_NAME
----------------------------------------
Kate Winslet
Kevin Spacey
Lana Condor
Maggie Gyllenhaal
Mark Wahlberg
Michael Cimino
Milos Forman
Nicole Kidman
Orson Welles
Paul Thomas Anderson
Peter Jackson

FULL_NAME
----------------------------------------
Peter OToole
Raoul Walsh
Richard Kelly
Ridley Scott
Robert De Niro
Robert Duvall
Robin Williams
Roman Polanski
Sam Mendes
Shelley Duvall
Sigourney Weaver
```

```
FULL_NAME
----------------------------------------
Stanley Kubrick
Stephen Baldwin
Susan Johnson
Tim Robbins
Woody Allen

60 rows selected.
```

4.Find the movie titles that did not receive any ratings.

Solution Approach

```sql
SELECT MOV_TITLE
FROM MOVIE
WHERE MOV_ID NOT IN (
    SELECT DISTINCT MOV_ID
    FROM RATING
);
```

SQL Command

```
MOV_TITLE
----------------------------------------------------
Citizen Kane
They Drive by Night
Spirited Away
To All the Boys I've Loved Before
The Theory of Everything
Back to the Future
Seven Samurai

7 rows selected.
```

5.Find the average rating of all movies.

Solution Approach

```
SELECT AVG(REV_STARS)
FROM RATING;
```

SQL Command

```
AVG(REV_STARS)
--------------
           6.8
```

6.Find the minimum rating for each movie and display them in descending order of rating.

Solution Approach

```
SELECT MOV_ID, MIN(REV_STARS) AS MIN_RATING
FROM RATING
GROUP BY MOV_ID
ORDER BY MIN_RATING DESC;
```

SQL Command

```
    MOV_ID MIN_RATING
---------- ----------
       933          7
       924          6
       917          6
       903          5
       909          5
       913          5
       916          5
       905          4
       901          4
       919          4
       902          4

    MOV_ID MIN_RATING
---------- ----------
       904          4
       914          3
       923          3
       920          3
       921          3
       918          3
       925          3
       906          3
       915          3
       910          3
       907          3

    MOV_ID MIN_RATING
---------- ----------
       911          3
       922          2
       908          2
       912          1

26 rows selected.
```

7.Find the title of the movie having an average rev_star higher than the average rev_star of all

the movies.

Solution Approach

```
SELECT MOV_TITLE
FROM MOVIE
WHERE MOV_ID IN (
SELECT MOV_ID
FROM RATING
GROUP BY MOV_ID
HAVING AVG(REV_STARS) > (SELECT AVG(REV_STARS) FROM
RATING));
```

SQL Command

```
MOV_TITLE
----------------------------------------
Deliverance
Amadeus
The Prestige
The Innocents
Good Will Hunting
The Shining
The Shawshank Redemption
Chinatown
Titanic
Eyes Wide Shut
Annie Hall

MOV_TITLE
----------------------------------------
Avatar
Braveheart

13 rows selected.
```

8.Find the name of actors/actresses and the number of ratings received by the movies in which

they played a role.

Solution Approach

```
SELECT a.ACT_FIRSTNAME || ' ' || a.ACT_LASTNAME AS
ACTOR_ACTRESS_NAME,
m.MOV_TITLE,
COUNT(r.MOV_ID) AS NUMBER_OF_RATINGS
FROM ACTOR a, CASTS c, MOVIE m, RATING r
```

```
WHERE a.ACT_ID = c.ACT_ID
AND c.MOV_ID = m.MOV_ID
AND m.MOV_ID = r.MOV_ID
GROUP BY a.ACT_FIRSTNAME, a.ACT_LASTNAME, m.MOV_TITLE
ORDER BY ACTOR_ACTRESS_NAME, NUMBER_OF_RATINGS DESC;
```

SQL Command

```
ACTOR_ACTRESS_NAME
----------------------------------------
MOV_TITLE                                          NUMBER_OF_RATINGS
-------------------------------------------------- -----------------
Christian Bale
The Prestige                                                       9

Claire Danes
Princess Mononoke                                                 10

Deborah Kerr
The Innocents                                                     13


ACTOR_ACTRESS_NAME
----------------------------------------
MOV_TITLE                                          NUMBER_OF_RATINGS
-------------------------------------------------- -----------------
Dev Patel
Slumdog Millionaire                                               10

Ewan McGregor
Trainspotting                                                     16

F. Murray Abraham
Amadeus                                                           13


ACTOR_ACTRESS_NAME
----------------------------------------
MOV_TITLE                                          NUMBER_OF_RATINGS
-------------------------------------------------- -----------------
Harrison Ford
Blade Runner                                                      13

Jack Nicholson
Chinatown                                                         11
```

```
James Stewart
Vertigo                                                                  9


ACTOR_ACTRESS_NAME
----------------------------------------
MOV_TITLE                                          NUMBER_OF_RATINGS
-------------------------------------------------- -----------------
Jon Voight
Deliverance                                                         11

Kate Winslet
Titanic                                                             15

Kevin Spacey
Beyond the Sea                                                      10


ACTOR_ACTRESS_NAME
----------------------------------------
MOV_TITLE                                          NUMBER_OF_RATINGS
-------------------------------------------------- -----------------
Kevin Spacey
American Beauty                                                     10

Maggie Gyllenhaal
Donnie Darko                                                        14

Mark Wahlberg
Boogie Nights                                                       12


ACTOR_ACTRESS_NAME
----------------------------------------
MOV_TITLE                                          NUMBER_OF_RATINGS
-------------------------------------------------- -----------------
Nicole Kidman
Eyes Wide Shut                                                      13
```

```
Peter OToole
Lawrence of Arabia                                              12

Robert De Niro
The Deer Hunter                                                 10


ACTOR_ACTRESS_NAME
----------------------------------------
MOV_TITLE                                          NUMBER_OF_RATINGS
-------------------------------------------------- -----------------
Robin Williams
Good Will Hunting                                               15

Shelley Duvall
The Shining                                                     10

Sigourney Weaver
Aliens                                                          16


ACTOR_ACTRESS_NAME
----------------------------------------
MOV_TITLE                                          NUMBER_OF_RATINGS
-------------------------------------------------- -----------------
Stephen Baldwin
The Usual Suspects                                              13

Tim Robbins
The Shawshank Redemption                                        13

Woody Allen
Annie Hall                                                      14


24 rows selected.
```

9.Find the name of the director of the movie having the highest average rev_star.

Solution Approach

```
SELECT d.DIR_FIRSTNAME || ' ' || d.DIR_LASTNAME AS
Director_Name
FROM DIRECTOR d
JOIN DIRECTION di ON d.DIR_ID = di.DIR_ID
```

```
JOIN (
SELECT MOV_ID, AVG(REV_STARS) AS Avg_Rev_Star
FROM RATING
GROUP BY MOV_ID
HAVING AVG(REV_STARS) = (
SELECT MAX(Avg_Rev_Star)
FROM (
SELECT MOV_ID, AVG(REV_STARS) AS Avg_Rev_Star
FROM RATING
GROUP BY MOV_ID)))
r ON di.MOV_ID = r.MOV_ID;
```

SQL Command

```
DIRECTOR_NAME
-------------------------
Stanley Kubrick
```

10.Find all the movie-related information of movies acted and directed by the same person.

Solution Approach

```
SELECT m.MOV_ID, m.MOV_TITLE, m.MOV_YEAR, m.MOV_TIME,
m.MOV_LANGUAGE, m.MOV_RELEASEDATE, m.MOV_COUNTRY
FROM MOVIE m, CASTS c, DIRECTION d
WHERE m.MOV_ID = c.MOV_ID
AND m.MOV_ID = d.MOV_ID
AND c.ACT_ID = d.DIR_ID;
```

SQL Command

```
no rows selected
```

11.Find the title and average rating of the movies that have an average rev_star of more than 7.

Solution Approach

```
SELECT m.MOV_TITLE, AVG(r.REV_STARS) AS
AVERAGE_RATING
FROM MOVIE m, RATING r
WHERE m.MOV_ID = r.MOV_ID
GROUP BY m.MOV_TITLE
HAVING AVG(r.REV_STARS) > 7
ORDER BY AVERAGE_RATING DESC;
```

SQL Command

```
MOV_TITLE                                          AVERAGE_RATING
-------------------------------------------------- --------------
The Shining                                                   8.4
The Shawshank Redemption                               8.23076923
Braveheart                                             7.54545455
Good Will Hunting                                      7.42857143
Avatar                                                 7.42857143
Chinatown                                                     7.4
Deliverance                                            7.27272727
Eyes Wide Shut                                         7.15384615

8 rows selected.
```

12. Find the reviewer who gives the highest number of lowest rev_star.

Solution Approach

```
SELECT r.REV_NAME, COUNT(*) AS Lowest_Rev_Star_Count
FROM REVIEWER r, RATING ra
WHERE r.REV_ID = ra.REV_ID
AND ra.REV_STARS = (
SELECT MIN(REV_STARS)
FROM RATING)
GROUP BY r.REV_NAME
HAVING COUNT(*) = (
SELECT MAX(Count_Rev_Star)
FROM (
SELECT COUNT(*) AS Count_Rev_Star
FROM REVIEWER r, RATING ra
WHERE r.REV_ID = ra.REV_ID
AND ra.REV_STARS = (
SELECT MIN(REV_STARS)
FROM RATING)
GROUP BY r.REV_NAME));
```

SQL Command

```
REV_NAME                    LOWEST_REV_STAR_COUNT
--------------------------- ---------------------
Scott LeBrun                                    1
```

13. Find the name and average runtime of movies of different actors/actresses. Do not include

any actor/actress who worked with ‘James Cameron’.

Solution Approach

```
SELECT (a.ACT_FIRSTNAME || ' ' || a.ACT_LASTNAME) AS
Actor_Actress_Name, AVG(m.MOV_TIME) AS
Average_Runtime
FROM ACTOR a, CASTS c, MOVIE m
WHERE a.ACT_ID = c.ACT_ID
AND c.MOV_ID = m.MOV_ID
AND a.ACT_ID NOT IN (
SELECT DISTINCT c.ACT_ID
FROM CASTS c, DIRECTION d, DIRECTOR di
WHERE c.MOV_ID = d.MOV_ID
AND d.DIR_ID = di.DIR_ID
AND (di.DIR_FIRSTNAME || ' ' || di.DIR_LASTNAME) =
'James Cameron')
GROUP BY a.ACT_FIRSTNAME, a.ACT_LASTNAME;
```

SQL Command

ACTOR_ACTRESS_NAME	AVERAGE_RUNTIME
Woody Allen	93
Robin Williams	126
Christian Bale	130
Eddie Redmayne	123
Lana Condor	99
Orson Welles	119
Mark Wahlberg	155
George Raft	95
James Stewart	128
Harrison Ford	117
Nicole Kidman	159

ACTOR_ACTRESS_NAME	AVERAGE_RUNTIME
Claire Danes	134
Tim Robbins	142
Kevin Spacey	120
Dev Patel	120
Jon Voight	109
Felicity Jones	123
Maggie Gyllenhaal	113
Deborah Kerr	100
Ewan McGregor	94
Peter OToole	216
Jack Nicholson	130

ACTOR_ACTRESS_NAME	AVERAGE_RUNTIME
Robert De Niro	183
F. Murray Abraham	160
Stephen Baldwin	106
Shelley Duvall	146

26 rows selected.

Problems faced :

Syntax Errors: SQL queries must have some strict syntax requirements. Incorrectly worded queries might result in syntax problems and as a beginner I did struggle to get the syntax properly.

Understanding Database Structure: As I was not so familiar with the structure of the database I am working with, it might be difficult to determine which tables exist, what columns they include, and how they relate to one another.

Table Creation: Creating tables with the right structure and data types were a bit tricky. Defining primary keys,**Foreign keys**, and constraints correctly is essential.

Connect Table : to connect multiple table command is a bit difficult to me.